

Supplementary material

Population genomics of the northern shortfin squid *Illex illecebrosus*

September 15, 2026

Supplementary figures

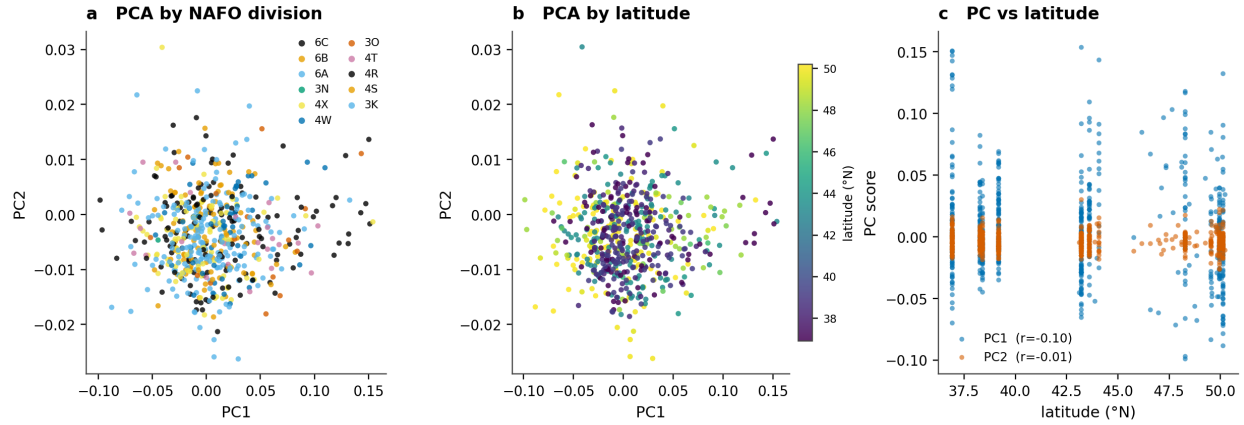


Figure S1: **The autosomal-PCA spread is not geographic structure.** Genome-wide PCA (autosomes; chromosome 2, chromosome 42 and the Z excluded), $n = 621$ individuals. (a) Coloured by NAFO sampling division and (b) by latitude, the divisions and the latitudinal gradient are fully intermixed across PC space — no clustering by origin. (c) PC scores plotted against latitude show no relationship (Pearson $r = -0.10$ for PC1, $r = -0.01$ for PC2). The dispersion along the minor PC axis in ?? is carried by a handful of individuals (~ 9 samples account for $>80\%$ of the PC2 variance) and tracks neither division nor latitude nor sequencing coverage; a KING-robust screen finds no cryptic relatedness among them, so it is a minor-axis artifact rather than population subdivision, consistent with genome-wide $F_{ST} \approx 0$.

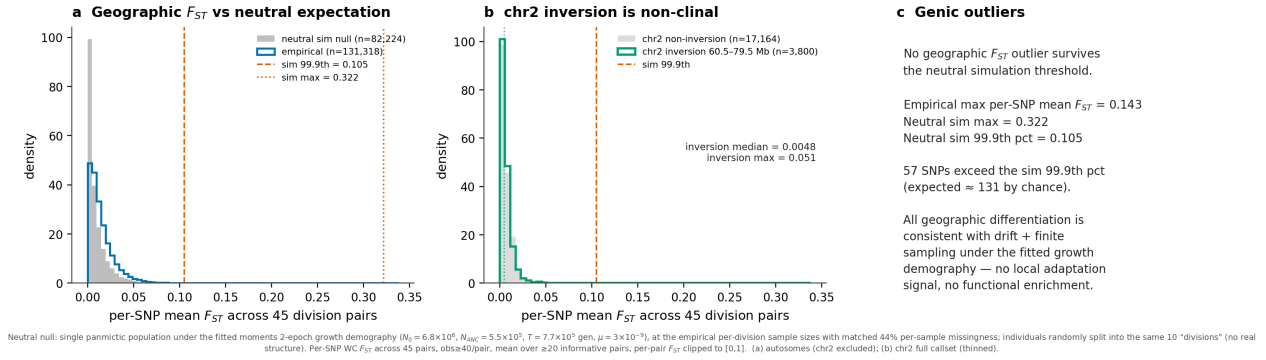


Figure S2: No geographic differentiation beyond genetic drift; no local adaptation signal. Per-SNP Weir–Cockerham F_{ST} across all 45 NAFO-division pairs (well-genotyped sites, ≥ 40 observed haplotypes per pair; per-SNP mean over ≥ 20 informative pairs). **(a)** The empirical distribution (blue) lies entirely within a coalescent *neutral null* (grey): a single panmictic population simulated under the fitted growth demography ($N_0 = 6.8 \times 10^6$, $N_{ANC} = 5.5 \times 10^5$, $\mu = 3 \times 10^{-9}$) at the empirical per-division sample sizes with the matched 44% missingness, split at random into ten “divisions”. Empirical max $F_{ST} = 0.143$ vs neutral max 0.322; only 57 SNPs exceed the neutral 99.9th percentile (0.105) against ≈ 131 expected by chance — *fewer* than the false-positive rate. Drift, finite sampling and missingness fully explain the observed differentiation. **(b)** On chromosome 2, the inverted region (green) is geographically indistinguishable from the collinear remainder (median $F_{ST} \approx 0.005$) — the inversion is non-clinal; its $F_{ST} \approx 0.37$ is between *karyotypes*, not geography. **(c)** No SNP survives the neutral threshold, so a scan for local adaptation that the diversity-based sweep tests could miss returns no candidates. (The autosomal panel samples a fixed 400 kb window per chromosome; the distribution/threshold conclusion is robust, but a genome-wide genic-outlier scan would require per-chromosome F_{ST} on the full callsets.)

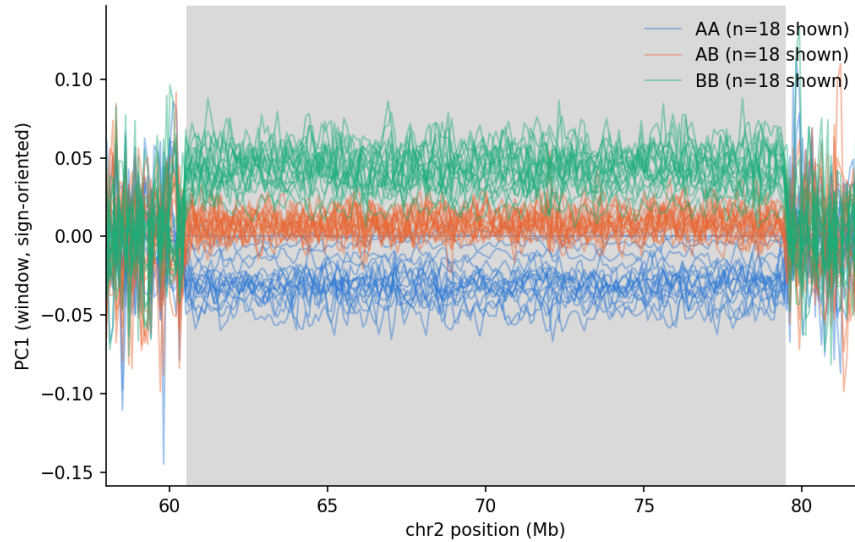


Figure S3: Windowed PCA across chromosome 2 localising the differentiated block. [Regeneration note: window coordinates to be corrected in the final version.]

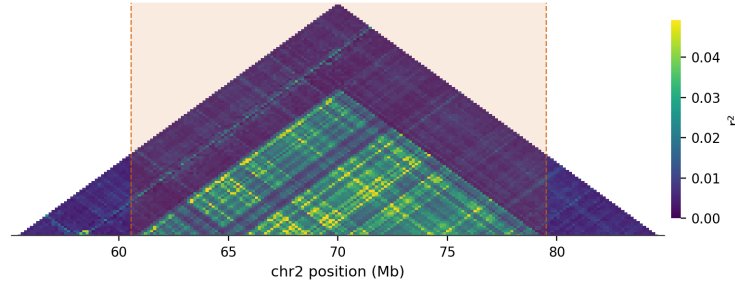


Figure S4: **Linkage-disequilibrium triangle across chr2**, showing the extended LD block coincident with the inversion.

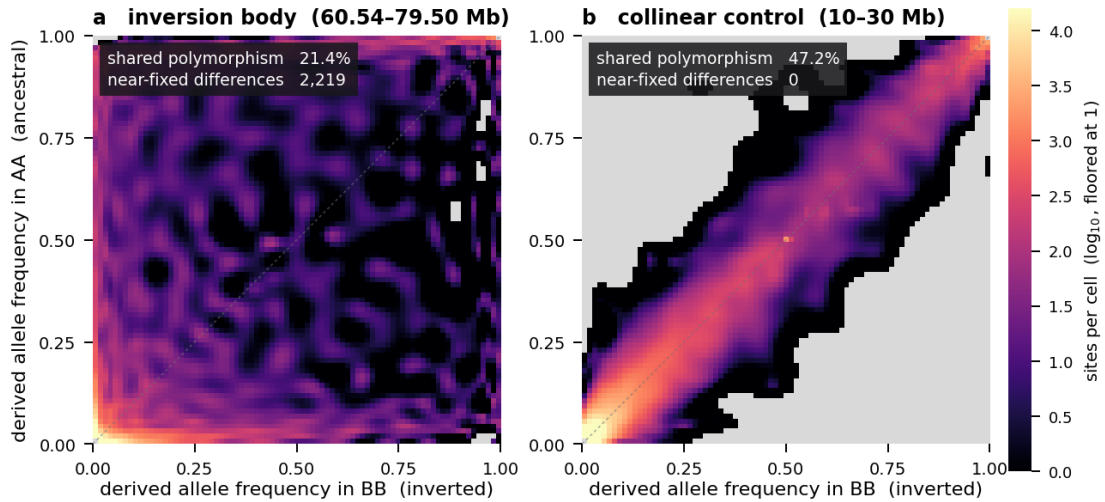


Figure S5: **Joint two-dimensional site-frequency spectrum** of derived-allele frequency in the AA (ancestral) versus BB (inverted) arrangement. Each cell is a count of sites (colour, $\log_{10}$). In the inversion body (a) density spreads off the diagonal and into the corners — variants common in one arrangement and absent in the other, i.e. near-fixed differences (2,219 sites; 21% shared polymorphism) — the direct signature of the recombination barrier. In the collinear control (b) the density collapses onto the diagonal (0 near-fixed differences; 47% shared), as expected for freely recombining, exchangeable sequence. *[Regeneration note: drop interpolation, floor counts at 1, mask empty cells to a distinct colour.]*

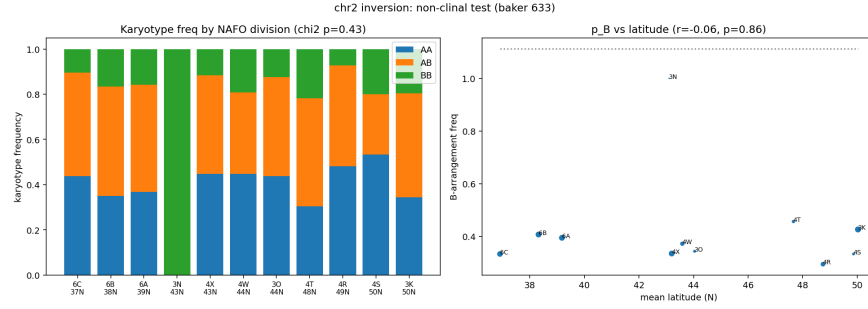


Figure S6: **The inversion is non-clinal.** Inverted-arrangement frequency across NAFO sampling divisions shows no latitudinal trend (see table S2); geographic $F_{ST} \approx 0$, and the three karyotypes co-occur throughout the range.

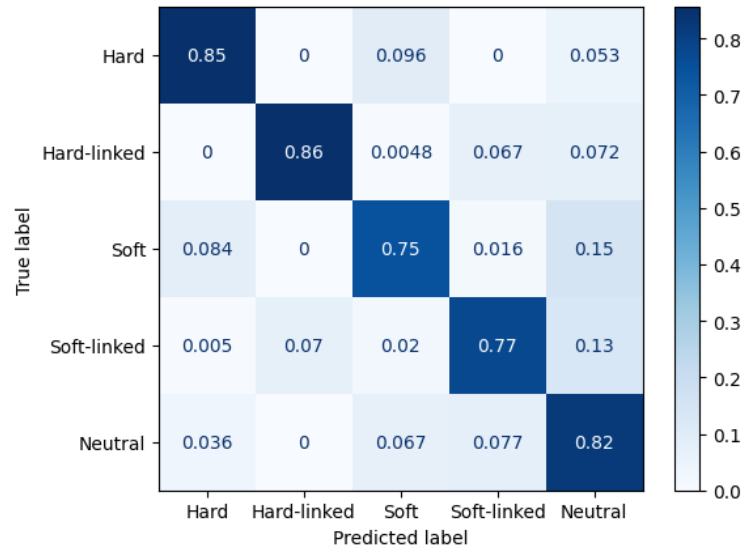


Figure S7: **Sweep-classifier confusion matrix** (five-class diploSHIC, full-SFS model; 81% overall accuracy). Recall: hard 0.85, linked-hard 0.86, neutral 0.82, soft 0.75, linked-soft 0.77.

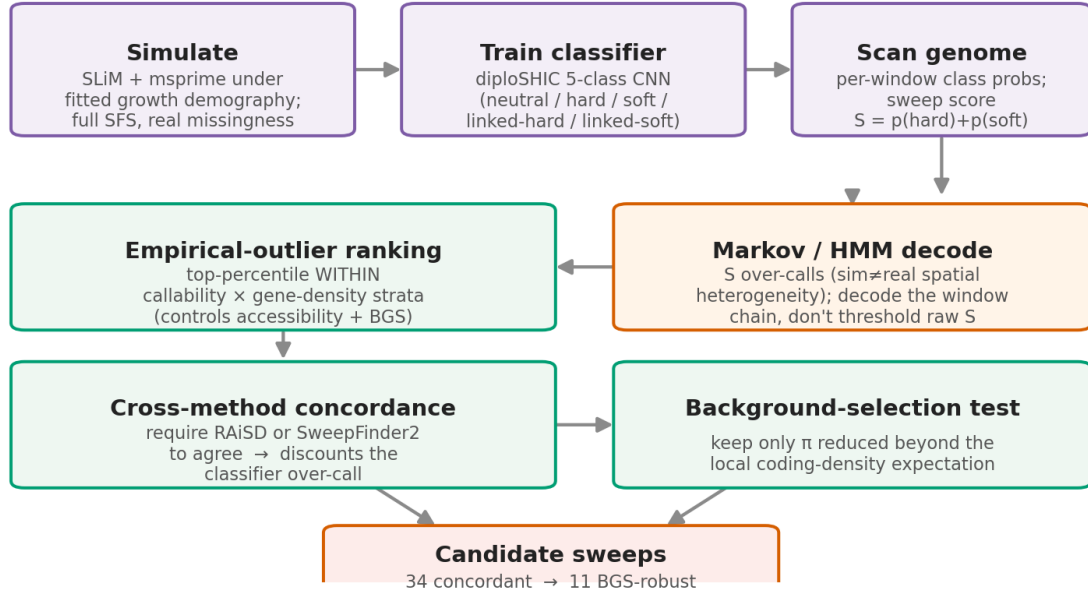


Figure S8: **The selective-sweep scan pipeline.** Simulate (SLiM + msprime under the fitted growth demography) → diploSHIC five-class classifier → Markov/HMM decode → stratified empirical-outlier ranking (within callability × gene-density strata) + cross-method concordance (RAiSD, SweepFinder2) + background-selection filter → candidate sweeps.

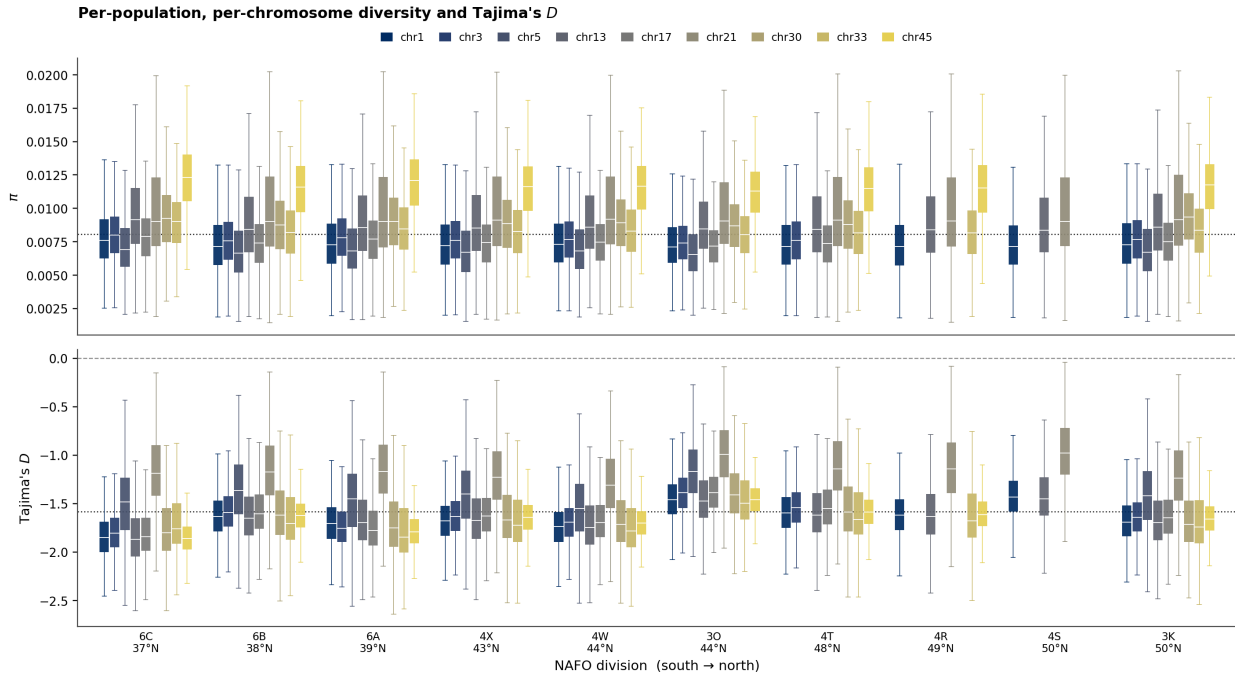


Figure S9: **Per-chromosome diversity.** Distribution of nucleotide diversity π (top) and Tajima's D (bottom) across windows, per chromosome. D is negative on every chromosome.

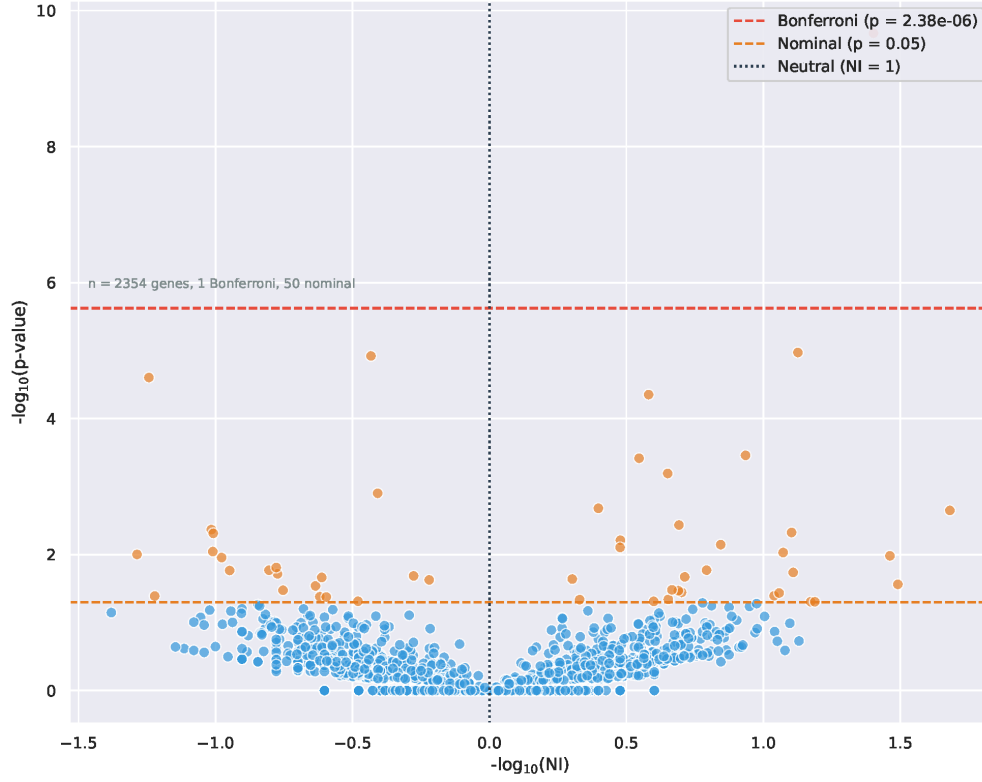


Figure S10: **Per-gene McDonald–Kreitman neutrality index** (mkado volcano plot, *I. coindetii* outgroup). Each point is one gene: $-\log_{10}$ of its neutrality index $NI = (P_n/P_s)/(D_n/D_s)$ on the x -axis (right of the neutral $NI = 1$ line = adaptive, $NI < 1$; left = excess non-synonymous polymorphism, $NI > 1$) against $-\log_{10} p$ on the y -axis. Dashed lines mark the nominal ($p = 0.05$) and Bonferroni thresholds. Of $n = 2,354$ testable genes, 50 are nominally significant and 1 survives Bonferroni; the genome-wide asymptotic $\alpha = 0.64$ resolves to few individual genes, as expected for a large- N_e species where per-gene MK is underpowered.

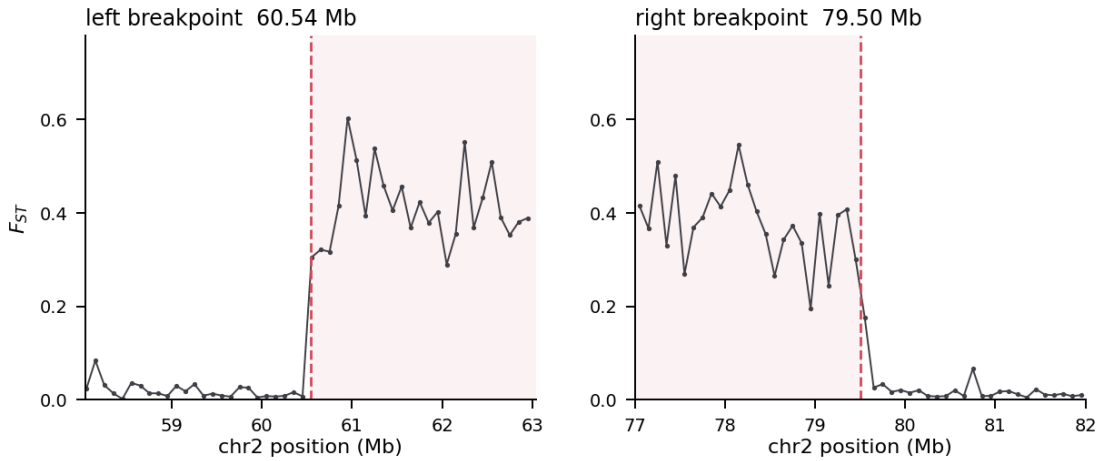


Figure S11: **Inversion breakpoints.** F_{ST} across the left (60.54 Mb) and right (79.50 Mb) breakpoints — a step, not a ramp.

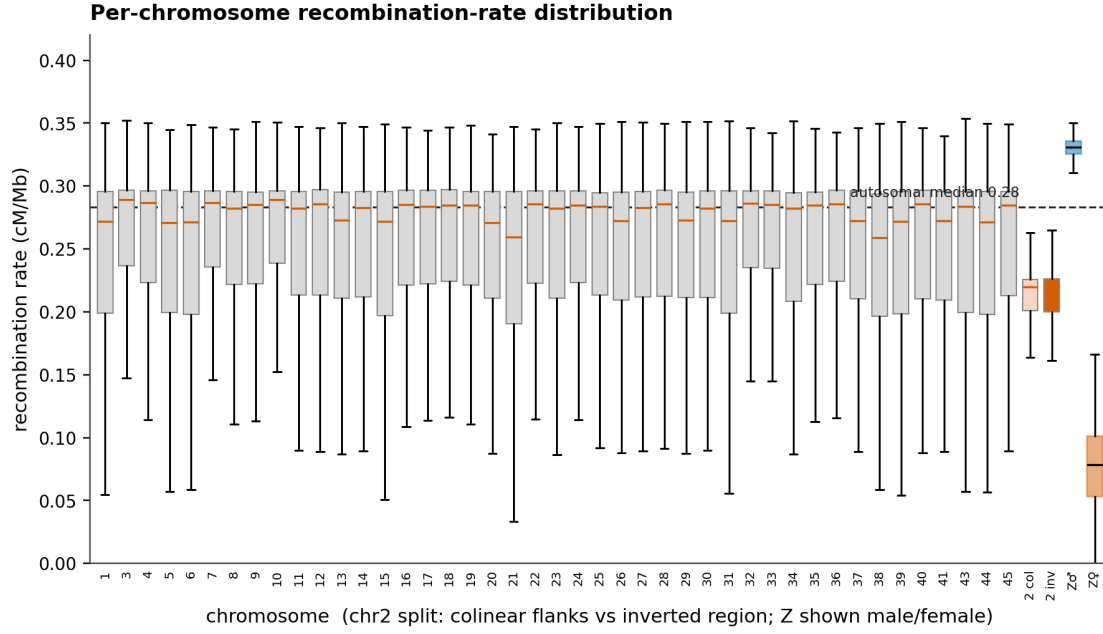


Figure S12: **Per-chromosome recombination-rate distribution** (ReLERNN). chr2 (autosomal-net estimate, flagged) and the Z chromosome (male/female) shown separately. *[chr2 and Z estimates are provisional — see main text.]*

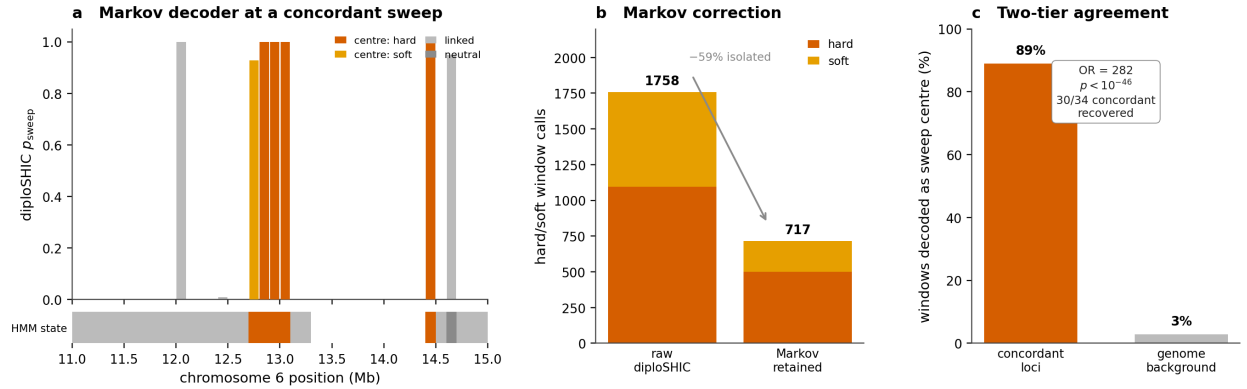


Figure S13: **Recombination-aware Markov decoder for the Tier-1 sweep calls.** A hidden Markov model over the per-100 kb-window diploSHIC classification (emissions = prior-corrected class posteriors; dwell lengths from the ~ 400 kb sweep footprint scaled by the ReLERNN map; design-informed, not simulation-calibrated). (a) The decoder at a representative concordant sweep (chromosome 6): the sweep posterior $p_{\text{hard}} + p_{\text{soft}}$ per window and the Viterbi state track (neutral / linked / centre); it localises the centre inside the linked envelope and overrides spatially isolated high-posterior windows (e.g. near 12.05 Mb). (b) Markov correction reduces the raw hard/soft calls from 1,758 to 717 spatially-coherent calls (-59%). (c) 89% of windows at the Tier-2 cross-method-concordant loci are decoded as sweep centres versus 2.9% genome-wide (odds ratio 282, $p < 10^{-46}$; 30 of 34 concordant regions recovered).

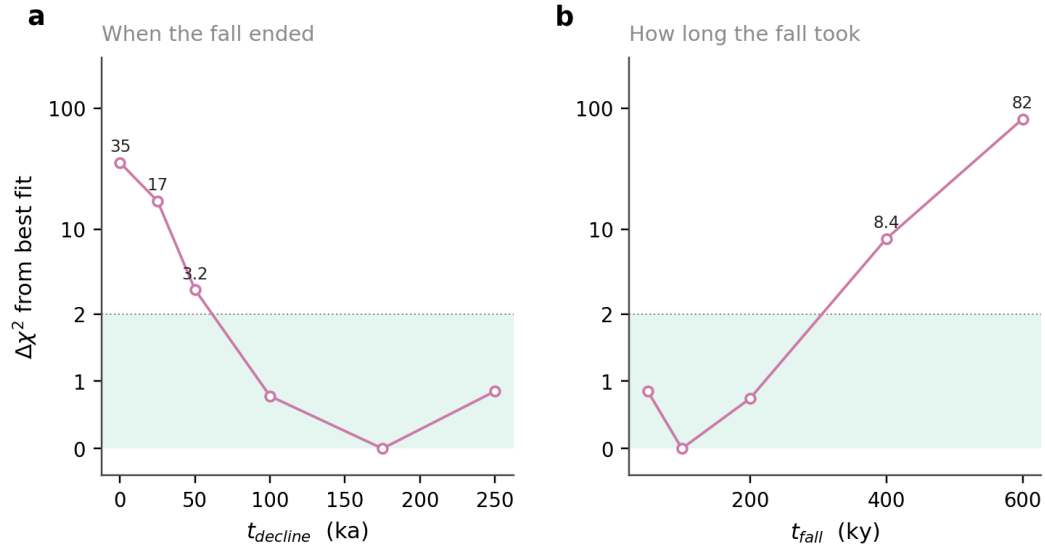


Figure S14: **Profile-likelihood bounds on the chromosome-2 inversion's decline (??)**. $\Delta\chi^2$ from the best fit while profiling (a) t_{decline} , when the fall ended, and (b) t_{fall} , how long it took; the shaded band is $\Delta\chi^2 < 2$ (not excluded). The fall ended $\sim 50\text{--}200$ ka and took $\sim 50\text{--}200$ ky.

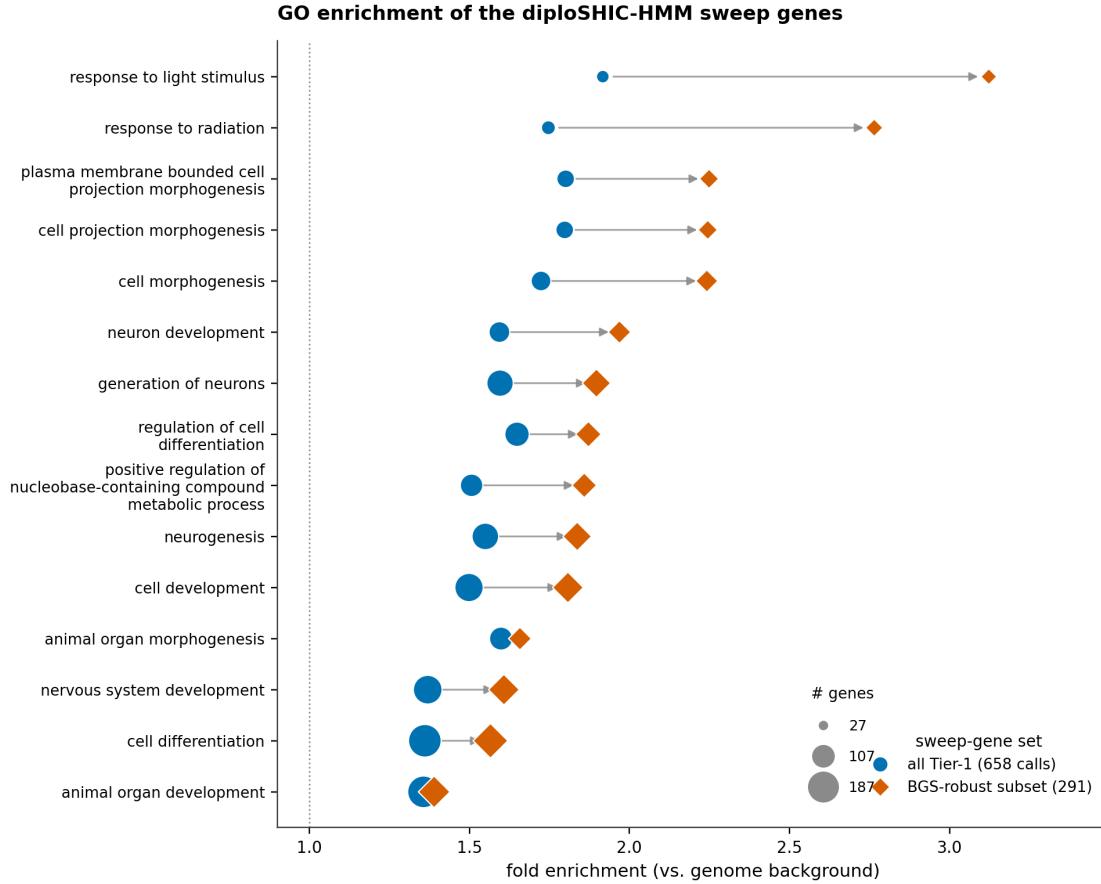


Figure S15: **GO enrichment of the diploSHIC-HMM sweep genes** (biological process; terms significant at $FDR < 0.05$ in both sets). Paired dot plot: each term is shown for the full Tier-1 set (blue circle; 658 calls) and for its background-selection-robust subset (orange diamond; 291 calls), with x = fold enrichment and marker area = number of sweep genes carrying the term. The grey arrow marks the shift between the two sets: the BGS screen consistently *sharpens* the enrichment (rightward), and most strongly for the light-response / photoreception terms (response to light stimulus $1.9 \rightarrow 3.1\times$; response to radiation $1.8 \rightarrow 2.8\times$). Two coherent themes dominate — neural development and photoreception — and both survive the background-selection filter (table S4; full per-set enrichment tables are provided with the source). The hard-sweep calls are a sub-classification recorded in the calls table, not a separate gene set.

BGS-robust sweep genes: GO enrichment map (shared-gene network)

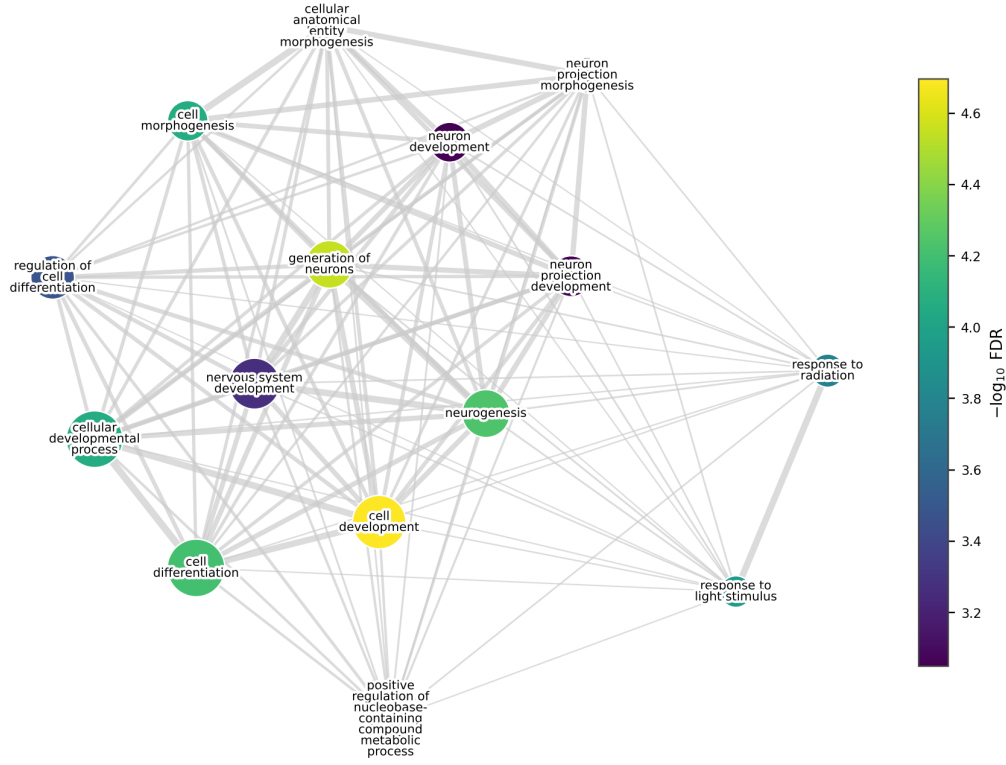


Figure S16: **GO enrichment map for the background-selection-robust sweep genes.** Built from the 291 BGS-robust Tier-1 calls (the subset whose local diversity reduction exceeds the background-selection expectation; 227 GO-annotated genes). Nodes are enriched biological-process terms (size = number of genes, colour = $-\log_{10}$ FDR); edges connect terms sharing genes (Jaccard ≥ 0.15). The terms form connected modules of neural development / cell morphogenesis and photoreception (response to light/radiation), the same two themes seen in the GSEA dot plot (fig. S15).

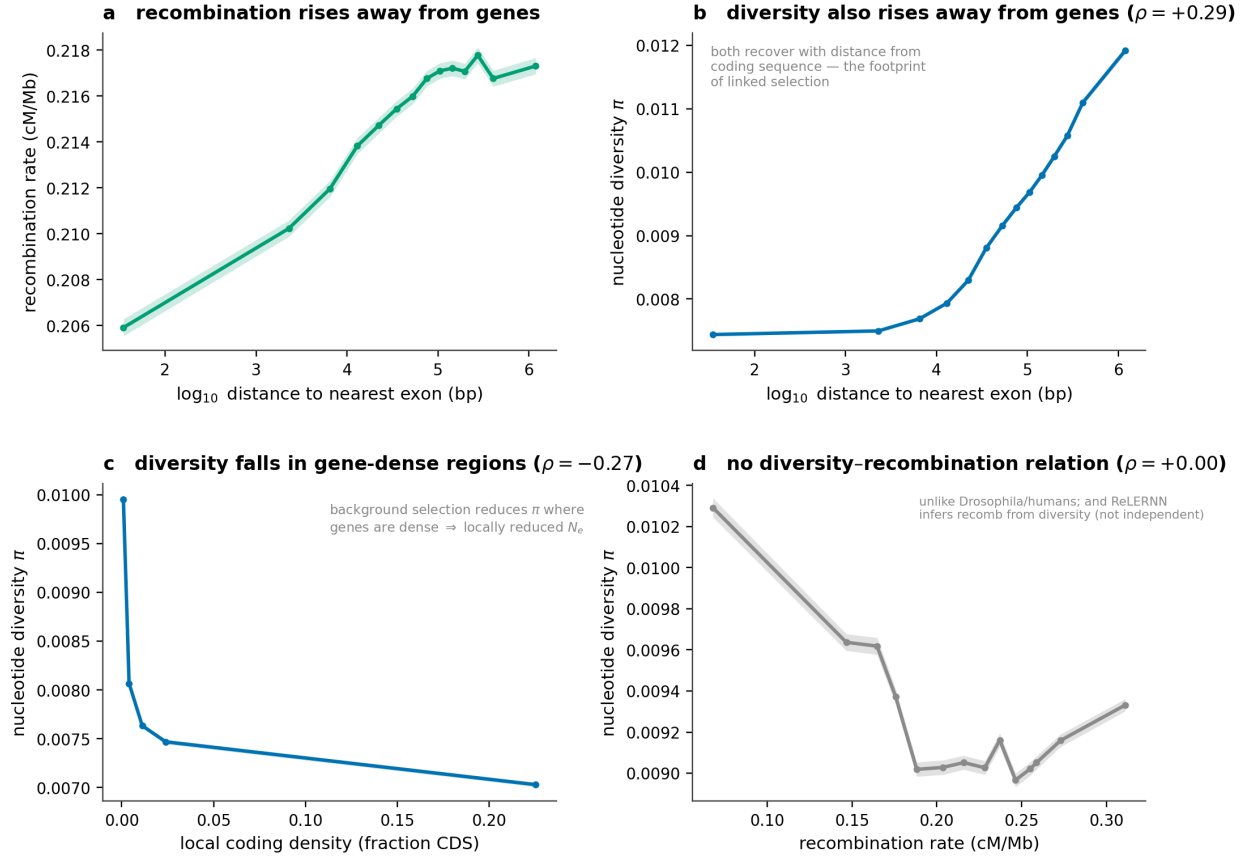


Figure S17: **Recombination, gene proximity and the efficacy of selection.** Per-window statistics (~ 50 kb windows, autosomes). **(a)** Recombination rate rises with distance from the nearest exon, and **(b)** nucleotide diversity π does the same ($\rho = +0.29$): gene proximity depresses both, the footprint of linked selection. **(c)** π falls with local coding density ($\rho = -0.27$) — background selection reduces diversity where genes are dense, i.e. a locally reduced effective size, and it is *gene-density-driven*. **(d)** There is *no* diversity–recombination relationship ($\rho \approx 0$), unlike *Drosophila* and humans; note also that ReLERNN infers recombination from diversity/LD, so π and the recombination estimate are not independent. Linked selection in *Illex illecebrosus* is thus organised by gene density and gene proximity rather than by the recombination-rate axis.

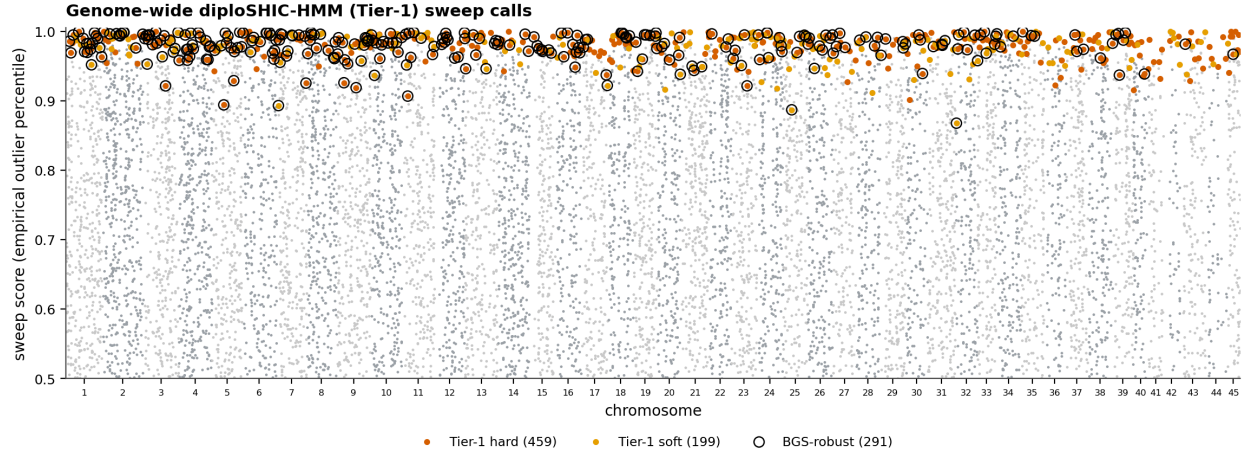


Figure S18: **Genome-wide diploSHIC-HMM (Tier-1) sweep calls.** Per-100 kb-window empirical-outlier sweep score (rank within callability $\times$ coding-density strata) across all 45 autosomes plus chromosomes 2 and 42. The Markov-decoded Tier-1 calls are highlighted by class (hard, soft), and the background-selection-robust subset (§ BGS screen; 291 of 658) is ringed. The calls are distributed across the genome with no single dominant locus.

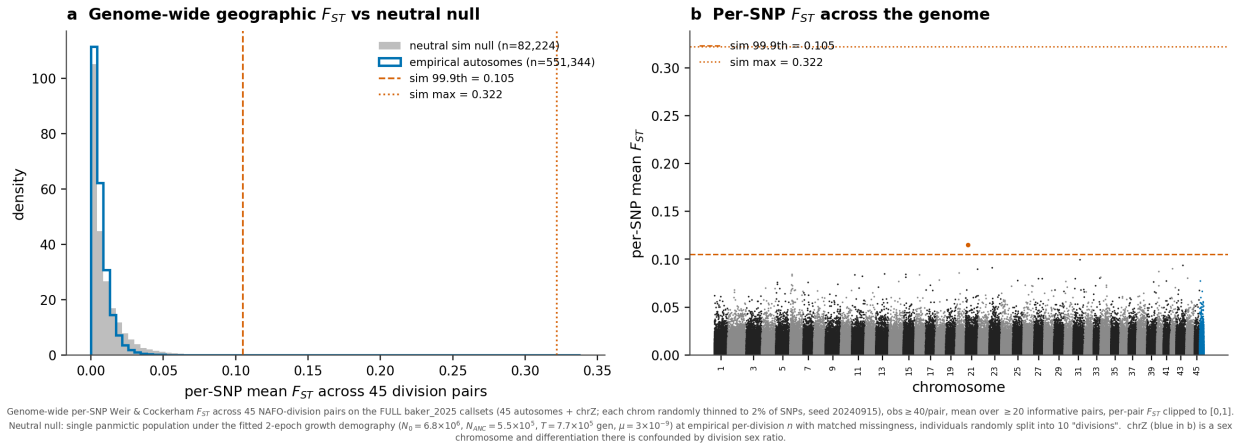


Figure S19: **Genome-wide per-SNP F_{ST} finds no differentiation outlier.** The window scan of fig. S2 extended to the *full* per-chromosome callsets (551,344 autosomal SNPs, 2% thinning, covering 12,007 genes). (a) The empirical per-SNP F_{ST} distribution across the 45 division pairs lies entirely within — and slightly narrower than — the coalescent neutral null. (b) Genome-wide, *no* SNP exceeds the neutral maximum (0.322); a single intergenic SNP on chromosome 21 exceeds the lenient 99.9th percentile (0.105), against ≈ 556 expected by chance — i.e. $\sim 555 \times$ *fewer* outliers than the false-positive rate. Zero genic outliers, hence no functional enrichment. Extreme panmixia is confirmed at genome scale, and no local-adaptation signal is present that the diversity-based sweep scans could have missed. (chrZ, blue in (b), is confounded by division sex ratios and shown for completeness.)

Table S1: **Software and versions.** Versions were read off the analysis machine (conda metadata, `pip list`, `--version`, or recorded `versions.txt`). Citations are pointers to confirm at final formatting.

Tool (version)	Used for	Reference
msinv 0.3.5	chr2 inversion age/decline (inversion)	(?); (?)
ANGSD 0.940	SFS, θ , diversity (demography, diversity)	(?)
Stairway Plot 2 (2.1.1)	demography	(?)
GONE2 1.0.4	recent demography	(?)
moments 1.6.0	demographic inference (demography)	(?)
demes 0.2.3	demography interchange	(?)
ReLERNN 0.2	recombination map	(?)
diploSHIC 1.1.0	sweep scan	(?)
RAiSD (0c705a8)	sweep corroboration (sweep scan)	(?)
SLiM 5.2	forward simulation	(?)
msprime 1.4.2	coalescent sim, mutation overlay	(?)
tskit 1.0.3	tree-sequence I/O	(?)
discoal 0.1.7a	sweep-sim validation	(?)
pg_gpu 0.1.1.dev14	windowed $\pi/d_{xy}/F_{ST}$, PCA	(?)
mkado 0.5.0	McDonald–Kreitman	(?)
DIAMOND 2.1.12	homology (inversion, sweep scan)	(?)
HMMER 3.4	Pfam domain screen	(?)
Pfam-A	domain screen	(?)
AnchorWave 1.2.3	<i>I. coindetii</i> ↔ <i>Illex illecebrosus</i> alignment	(?)
minimap2 2.29	alignment	(?)
est-sfs 2.04	ancestral polarisation	(?)
degenotate 1.3	0-/4-fold degeneracy	(?)
bedtools 2.30.0	interval operations	(?)
bcftools/samtools 1.21	VCF/BAM handling	(?)
EarlGrey 7.0.1	TE annotation	(?)
StringTie 3.0.0	transcript assembly	(?)
SQANTI3	Iso-Seq QC	(?)

Supplementary tables

Table S2: **Chromosome-2 karyotype counts and inverted-arrangement frequency by NAFO division** (ordered by latitude). The inverted (BB) frequency p_{inv} shows no latitudinal cline (range 0.29–0.46), consistent with geographic $F_{\text{ST}} \approx 0$ (fig. S6).

Division	AA	AB	BB	N	p_{inv}	mean lat. ($^{\circ}\text{N}$)
6C	38	40	9	87	0.333	36.9
6B	34	47	16	97	0.407	38.3
6A	35	45	15	95	0.395	39.2
3N	0	0	1	1	1.000	43.1
4X	42	41	11	94	0.335	43.2
4W	21	17	9	47	0.372	43.6
3O	7	7	2	16	0.344	44.0
4T	7	11	5	23	0.457	47.7
4R	27	25	4	56	0.295	48.7
4S	8	4	3	15	0.333	49.9
3K	35	47	20	102	0.426	50.0
Total	254	284	95	633	0.374	—

Table S3: **Use of the two outgroups.** *I. coindetii* served two roles — ancestral-state polarization of the chromosome-2 inversion and the divergence term of the genome-wide McDonald–Kreitman (MK) test — both resting on a single whole-genome *I. coindetii*↔*Illex illecebrosus* AnchorWave alignment. *I. argentinus* was aligned identically (CDS) and used as a secondary MK outgroup; it yields $\sim 9\times$ fewer fixed coding differences than *I. coindetii* yet has genome-wide π indistinguishable from *Illex illecebrosus*, marking it as a recently-diverged congener of near-identical diversity.

Step	Quantity	Value
<i>Alignment</i> (AnchorWave v1.2.3, CDS-anchored)		
	<i>I. coindetii</i> substitution sites (genome-wide)	45,722,335
	aligned scaffold identity (inversion body)	99.2–99.9%
<i>Polarization</i> of chr2 karyotypes (est-sfs 2.04 + non-circular subset)		
	diagnostic (AA/BB fixed-difference) sites	13,998
	... where <i>I. coindetii</i> = <i>Illex illecebrosus</i> REF (circular; excluded)	9,680 (69.2%)
	... where <i>I. coindetii</i> carries the ALT (non-circular; used)	4,175
	polarity from non-circular set	BB derived / AA ancestral
<i>McDonald–Kreitman</i> test (mkado, <i>I. coindetii</i> divergence)		
	genes tested	20,999
	nonsynonymous / synonymous divergence (Dn / Ds)	15,343 / 13,670
	nonsynonymous / synonymous polymorphism (Pn / Ps)	468,124 / 630,949
	asymptotic α (95% CI)	0.642 (0.539–3.673)
<i>I. argentinus</i> (secondary MK outgroup; CDS alignment)		
	nonsynonymous / synonymous divergence (Dn / Ds)	1,355 / 1,805
	fixed coding differences (vs <i>I. coindetii</i> 29,013)	3,160 ($\sim 9\times$ fewer)
	asymptotic α	0.526
	genome-wide π (ANGSD, 6 autosomes; <i>Illex illecebrosus</i> = 0.0093)	0.0092

Table S4: **Gene-ontology enrichment of the Tier-1 (diploSHIC-only, Markov-corrected) sweep genes.** The 658 Tier-1 calls overlap 894 genes (476 GO-annotated); enrichment by Fisher’s exact test against the annotated-gene background on the scanned chromosomes, Benjamini–Hochberg FDR. 124 biological-process terms reach $\text{FDR} < 0.05$ — representative terms shown, spanning the two dominant themes (neural development; RNA-regulatory/transcriptional control). All are carried by the hard-sweep calls; the soft-sweep gene set has no term at $\text{FDR} < 0.05$. Full machine-readable table: `hmm_decode/tier1_GO_enrichment.tsv`.

GO id	term (biological process)	genes (T1/bg)	fold	FDR
GO:0048699	generation of neurons	124/1444	1.60	1.5×10^{-4}
GO:0022008	neurogenesis	126/1511	1.55	1.6×10^{-4}
GO:0030182	neuron differentiation	91/1076	1.57	1.5×10^{-3}
GO:0030154	cell differentiation	187/2553	1.36	3.7×10^{-4}
GO:0009887	animal organ morphogenesis	96/1116	1.60	8.3×10^{-4}
GO:0050767	regulation of neurogenesis	67/695	1.79	9.4×10^{-4}
GO:0000902	cell morphogenesis	72/776	1.72	1.3×10^{-3}
GO:0007409	axonogenesis	45/463	1.81	1.0×10^{-2}
GO:0048858	cell projection morphogenesis	59/610	1.80	1.6×10^{-3}
GO:0051252	regulation of RNA metabolic process	145/1922	1.40	1.3×10^{-3}
GO:2001141	regulation of RNA biosynthetic process	135/1770	1.42	1.5×10^{-3}

Table S5: **Functional clusters of the background-selection-robust Tier-1 sweep genes.** The 152 enriched biological-process GO terms ($\text{FDR} < 0.05$) of the BGS-robust set (fig. S15), grouped into functional clusters by term semantics. Neural development and cell morphogenesis/differentiation dominate, with RNA-regulatory control and organ development, and a distinct photoreception/sensory cluster.

functional cluster	terms	fold range	representative term (min FDR)
Neural development	22	1.6–22.2	generation of neurons (3×10^{-5})
Cell morphogenesis & differentiation	17	1.5–2.2	cell development (2×10^{-5})
RNA / transcription regulation	14	1.5–2.1	reg. RNA metabolic process (9×10^{-4})
Organ / anatomical development	13	1.2–2.5	system development (3×10^{-3})
Photoreception / sensory response	7	1.2–5.8	response to light stimulus (1×10^{-4})
Other regulation / metabolism	65	1.2–22.2	reg. developmental process

Table S6: **Tier-2 (cross-method-concordant) sweep genes.** The 34 regions where the diploSHIC-HMM sweep signal is independently corroborated by both RAiSD and SweepFinder2 (33 auto-detected plus the chr43 balancing locus handled separately). Seven regions carry a named protein-coding gene (12 genes, listed); the remaining concordant regions are gene deserts or uncharacterised loci. Coordinates in Mb; α and DoS from the McDonald–Kreitman test against *I. coindetii* (‘–’ where the test is unpowered, i.e. too few fixed/polymorphic sites). The full candidate set — all 216 windows, the 34 concordant regions, and the background-selection test — is provided as machine-readable tables (`outlier_scan/{windows,regions,concordant33_genes_ENTAP_G0,concordant33_genes_MK,concordant33_genes_F`

region (Mb)	sweep	gene	EnTAP best hit	α	DoS
1:20.3	hard	<i>Aplnr</i>	Apelin receptor	1.00	+0.17
1:85.5	hard	<i>eif3h</i>	Eukaryotic translation initiation factor 3	0.69	+0.20
1:85.5	hard	<i>babam1</i>	BRISC and BRCA1-A complex member 1	–	-0.48
1:85.5	hard	<i>ttc39b</i>	Tetratricopeptide repeat protein 39B	–	-0.27
5:38.0	hard	<i>cept1</i>	Choline/ethanolaminephosphotransferase 1	–	-1.00
5:38.0	hard	<i>Arsi</i>	Arylsulfatase I	–	-0.25
13:73.2	hard	<i>Rab3</i>	Ras-related protein Rab-3	–	-0.17
13:73.2	hard	<i>hyi</i>	Putative hydroxypyruvate isomerase	–	-0.64
13:73.2	hard	<i>Vps28</i>	Vacuolar protein sorting-associated protein 28	–	–
32:9.3	soft	<i>MacroD2</i>	ADP-ribose glycohydrolase MACROD2	–	-0.59
33:9.0	hard	<i>Tinagl1</i>	Tubulointerstitial nephritis antigen-like	–	-0.23
35:39.0	hard	<i>zag-1</i>	Zinc finger E-box-binding homeobox (ZEB)	–	-0.31